Postgraduate School
University of Ibadan
Ibadan, Oyo State
Nigeria

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UK Visas & Immigration
Global Talent Visa — Tech Nation (Digital Technology)
United Kingdom

To Whom It May Concern,

I write to endorse Odeyemi Olusegun Israel for the UK Global Talent Visa under the Digital Technology route. I am a Professor of Electrical and Electronic Engineering at the University of Ibadan, where I currently serve as Provost of the Postgraduate School. My research spans power systems, computational intelligence, and applied machine learning. I have mentored Olusegun for over 10 years and have reviewed his technical work in detail.

AMTTP: a production compliance platform built by one person.

Olusegun has designed and built the Anti-Money Laundering Transaction Trust Protocol (AMTTP)—a four-layer system comprising client SDKs (TypeScript and Python, both open-sourced), a compliance orchestration engine of 7 parallel microservices, an ML training pipeline processing 2.64 million Ethereum transactions (production ROC-AUC of 0.9879), and 18 Solidity smart contracts on Ethereum Sepolia. The whole stack runs across 17 containerised microservices.

AMTTP solves a problem the blockchain compliance industry has not yet solved: *pre-settlement* AML enforcement. Every major tool on the market—Chainalysis, Elliptic, TRM Labs—operates after a transaction has settled. AMTTP intervenes before settlement, completing ML inference, graph analysis, sanctions screening, and policy evaluation within Ethereum's 12-second block window. That is genuine real-time systems engineering.

He also built zkNAF, a zero-knowledge proof framework (Groth16 ZK-SNARKs on BN254) that lets institutions prove compliance facts—KYC validity, risk-score range, sanctions non-membership—without revealing personal data, resolving the GDPR-versus-AML conflict at the cryptographic level.

In a university, a project of this scope would require a PI, several postdocs, and a team of PhD students. Olusegun built it alone. I have reviewed the GitHub repository: there is no evidence of collaborators or outsourced components.

Blind-Spot Decomposition and the Geometry of System Collapse: a contribution I can assess directly.

Olusegun's research papers on Blind Spot Decomposition Theory (BSDT) and the Universal Detection Principle (UDP) are solid contributions to anomaly detection. But it is his third paper, *Blind-Spot Decomposition and the Geometry of System Collapse*, that I endorse most directly, because it contains results in power systems that fall squarely within my area of expertise.

In February 2021, Winter Storm Uri caused cascading blackouts across the Texas ERCOT grid. The public narrative blamed frozen wind turbines. Olusegun's adaptive friction framework, applied to ERCOT dispatch data, achieves:

- **Detection 72 hours before rolling blackouts began** (AUC = 0.9997).
- **Adaptive friction reduces peak capacity loss by 71%.**
- Per-agent BSDT decomposition identifies **gas supply chain disruption**—not frozen wind turbines—as the structural vulnerability driving the cascade.

The gas supply finding is exactly what the subsequent FERC/NERC joint investigation confirmed—but Olusegun's framework identifies it *mathematically, 72 hours in advance, from dispatch data alone*. As a power systems researcher, I can say that a 72-hour predictive lead on a grid cascade event, with correctly identified root cause, is an exceptional result. I am not aware of any other framework in the literature that achieves this.

The same paper validates on FDIC banking data (6-quarter lead on the 2008 GFC, zero false alarms), the Terra/Luna collapse ($r = 0.996$ correlation), and 3D Navier-Stokes fluid dynamics. That the same mathematical framework diagnoses blockchain fraud, predicts banking crises, and identifies power grid cascade origins is remarkable.

Relevance to the UK and my assessment.

The FCA's forthcoming crypto-asset regulatory framework will create demand for protocol-level compliance infrastructure. AMTTP is, as far as I am aware, the only system providing pre-settlement AML enforcement on blockchain transactions. The ERCOT and Navier-Stokes work also aligns with UK priorities in infrastructure resilience (Ofgem, National Grid ESO) and applied mathematics (EPSRC).

In my years at the University of Ibadan, I have mentored hundreds of engineers. Olusegun's engineering capability and systems-level thinking place him in the top tier of talent I have encountered. He designs and builds complex systems from first principles—across ML, blockchain, cryptography, and mathematical research—entirely without institutional support.

I recommend him for the Global Talent Visa without reservation.

Yours faithfully,

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